Exp.no: 13
DATE : 15/10/26

IMPLEMENT YOUR OWN PING PROGRAM

AIM:

To implement our own ping program and verify the reachability of a destination host

PROCEDURE:

1) Open VS code and create a Python file named my-ping.py
2) Import the required modules for socket communication & time measurement.
3) Get the destination IP address or hostname from the user.
4) Create a socket for communicating with the destination host.
5) Record the starting time before sending the request.
6) Send a request to the destination host.
7) Wait for the response from the destination.
8) Record the time when the response is received.
9) Calculate the response time.
10) Display the destination address and response time.

Explorer

13

my_ping.py

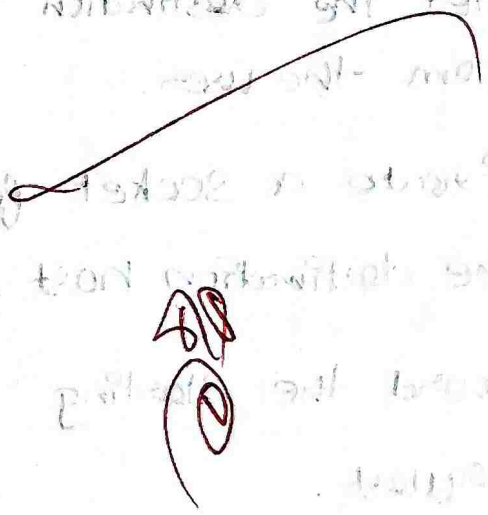
Welcome | my_ping.py

my_ping.py > ...

```
1   import socket
2   import time
3   host = input("Enter destination IP: ")
4   for i in range(4):
5       start = time.time()
6       try:
7           socket.gethostbyname(host)
8           end = time.time()
9
10          print("Reply from", host, "time =", round((end - start)
11
12      except:
13          print("Request timed out")
14      time.sleep(1)
```

Problems Output Terminal ... powershell

```
PS D:\13> python my_ping.py
Enter destination IP: 8.8.8.8
Reply from 8.8.8.8 time = 61.0 ms
Reply from 8.8.8.8 time = 0.0 ms
Reply from 8.8.8.8 time = 0.0 ms
Reply from 8.8.8.8 time = 0.0 ms
PS D:\13>
```



11) Repeat the ping operation several times to verify connectivity.

12) If the destination does not respond within the specified time, display a timeout message.

13) Run the program using the VS Code terminal:

```
python my-ping.py
```

14) Enter the destination IP address or hostname.

15) Observe the replies and response time.

16) Verify whether the destination host is reachable.

RESULT:

Thus the ping program was successfully implemented to verify the reachability of a destination host & measure its response time.